Quadric surfaces

degree 2

$f(x,y,z) = 0$

Cone

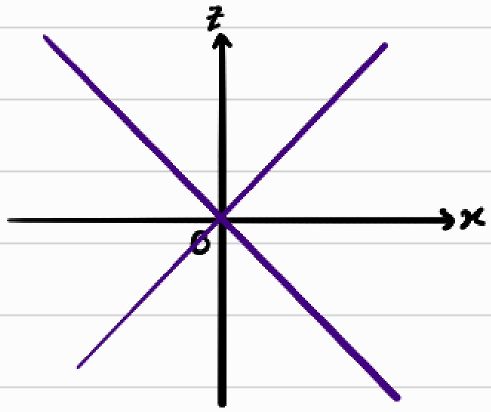
$x^2 + y^2 = z^2$

By observation, this shape consists of circles with radius z and height z that appear parallel to the

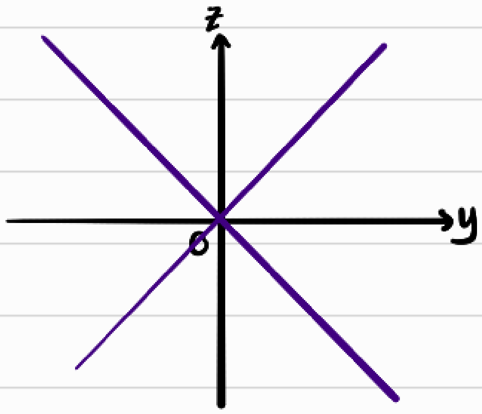
z = const.

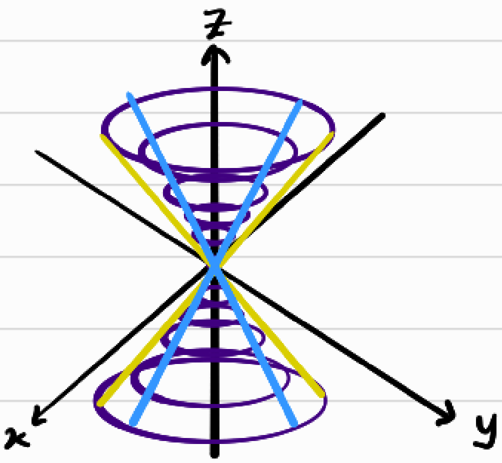
xy plane. Thus as height increase in magnitude, radius increases

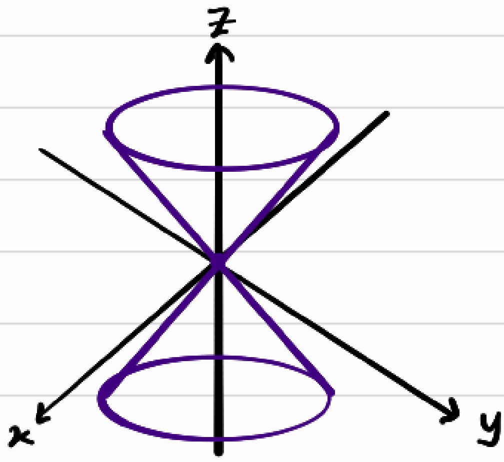
xz plane $y=0 \Rightarrow z = \pm x$



yz plane $x=0 \Rightarrow z = \pm y$





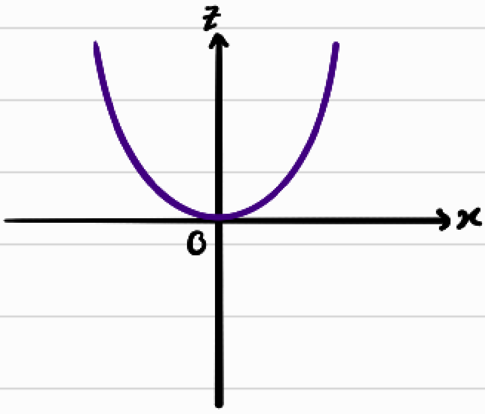


Parabloid

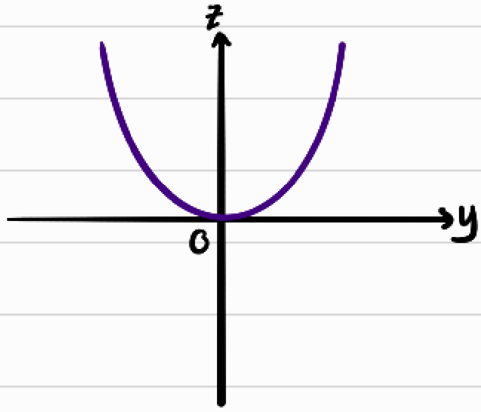
$$x^2 + y^2 = z$$

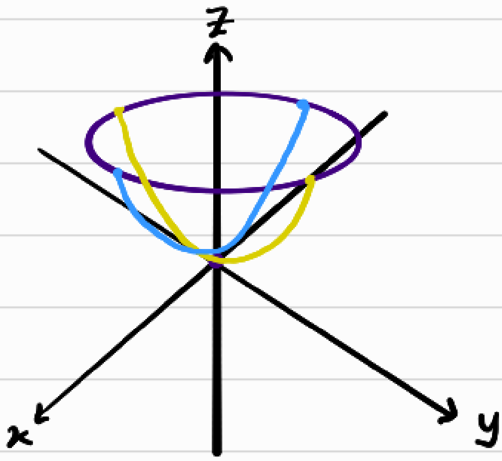
From observation, parallel to xy plane (z = constant), circles $\bar{w}$ increasing radius at increasing height are drawn AND $z > 0$

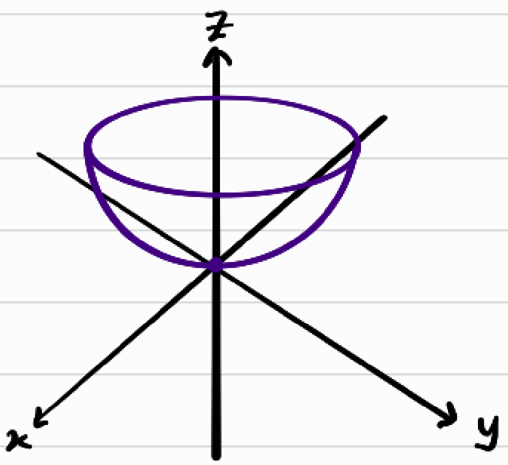
xz plane $y = 0 \Rightarrow z = x^2$



yz plane $x = 0 \Rightarrow z = y^2$



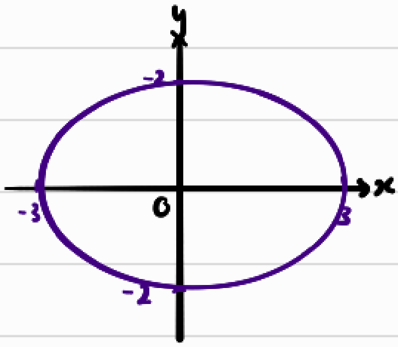




$\boxed{\text{Ellipsoid}}$ $\left(\frac{x}{3}\right)^2 + \left(\frac{y}{2}\right)^2 + z^2 = 1$

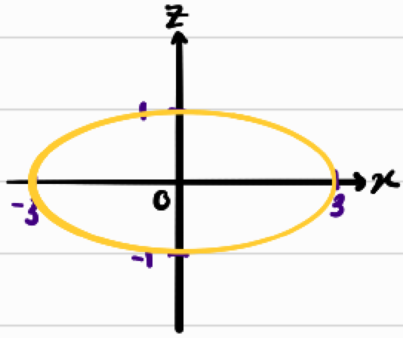
xy plane

$z = 0 \Rightarrow \left(\frac{x}{3}\right)^2 + \left(\frac{y}{2}\right)^2 = 1$



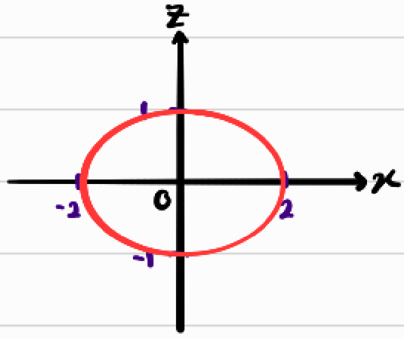
xz plane

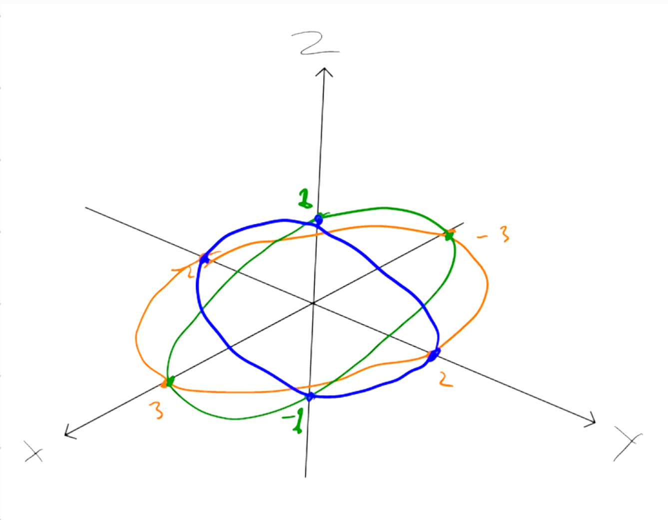
$y = 0 \Rightarrow \left(\frac{x}{3}\right)^2 + z^2 = 1$

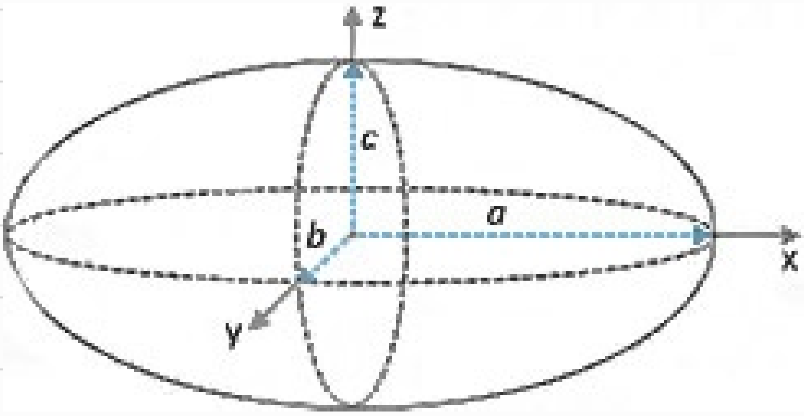


yz plane

$x = 0 \Rightarrow \left(\frac{y}{2}\right)^2 + z^2 = 1$





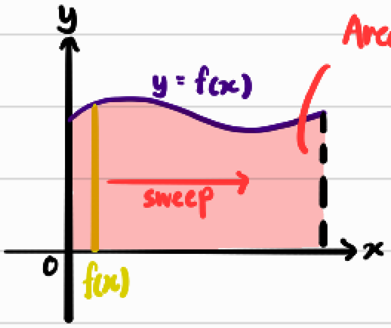


Triple Integration

Concept

Single Integration : $\int f(x)\,dx$

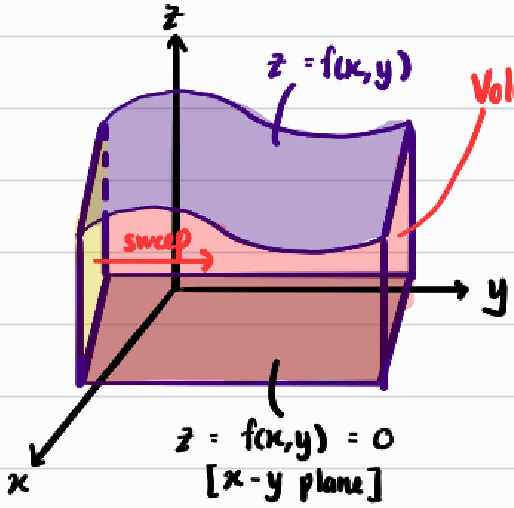
x = width , $f(x)$ = height



Area btwn f(x) & x-axis [y = 0]

Double Integration $\int \left[\int f(x,y)\,dx \right] dy$

x = width , y = length , $f(x,y)$ = height



Volume btwn f(x,y) & x-y plane [f(x,y) = 0]

Triple Integration $\iiint f(x,y,z)\,dx\,dy\,dz$

x = width, y = length, z = height,
$f(x,y,z)$ = a certain scalar property

Double Integration gets the "pure volume" of a 3D shape, ignoring any "stuff" inside it. What if we wanted to calculate the ammount of stuff inside a particular volume?

Double Integration essentially involves 2 steps

1) Sweep a "pure line" across direc. x to get "pure area"
2) Sweep the "pure area" across direc. y to get "pure volume"

Triple Integration essentially involves 3 steps

1) Sweeps the scalar field ($f(x,y,z)$) across direc. x to get a "weighted line"

[This "weighted line" gives us the sum of scalar properties of the scalar field in the direc. x (For any constant y or z)]

2) Sweep the "weighted line" across direc. y to get "weighted area"
3) Sweep the "weighted area" across direc. z to get "weighted volume"

} Double Integral

How much "stuff" is in this volume

Change of order

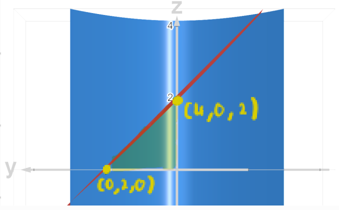
Example 1:

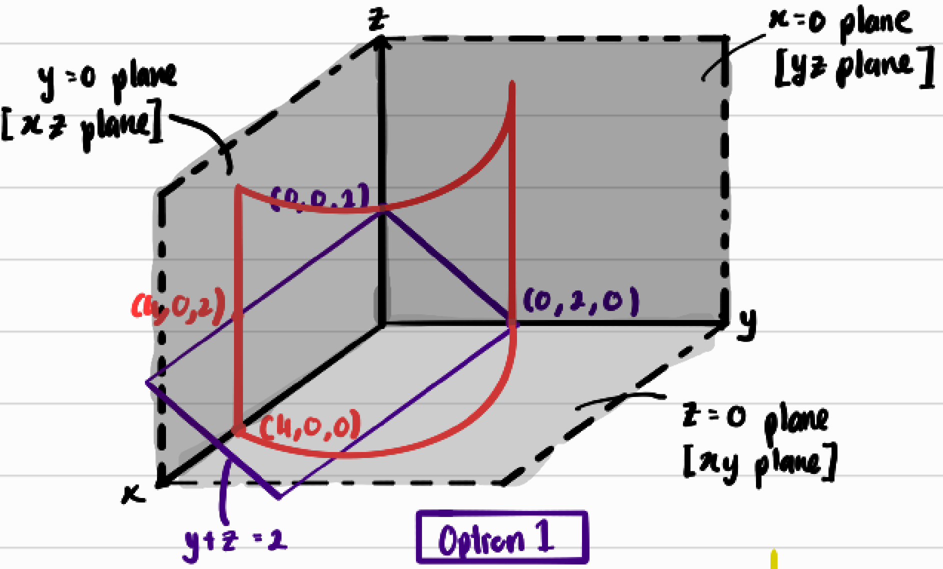
Consider the triple integral

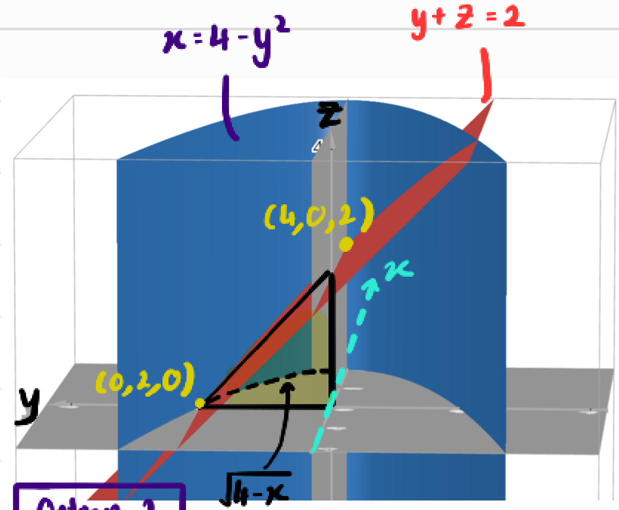
$f(x,y,z) = y$

$$\iiint_V y \, dV$$

over the 3D region bounded by $x=0$ plane, $y=0$ plane, $z=0$ plane, $y+z=2$ plane and parabolic cylinder $x = 4 - y^2$







Option 1

$0 \le z \le 2 - y$

$0 \le y \le \sqrt{4-x}$ sweep

$0 \le x \le 4$ sweep

we don't want to int. surd

$$\therefore \int_0^4 \int_0^{\sqrt{4-x}} \left(\int_0^{2-y} y \, dz \right) dy \, dx$$

Option 2

$0 \le z \le 2 - y$

$0 \le x \le 4 - y^2$ sweep

$0 \le y \le 2$ sweep

$$\therefore \int_0^2 \int_0^{4-y^2} \left(\int_0^{2-y} y \, dz \right) dx \, dy$$

$$= \int_0^2 \int_0^{4-y^2} 2y - y^2 \, dx \, dy$$

$$= \int_0^2 (8y - 2y^3 - 4y^2 + y^4) \, dy$$

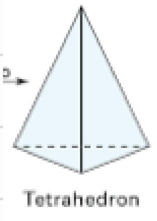
$$= 56/15 \;\#$$

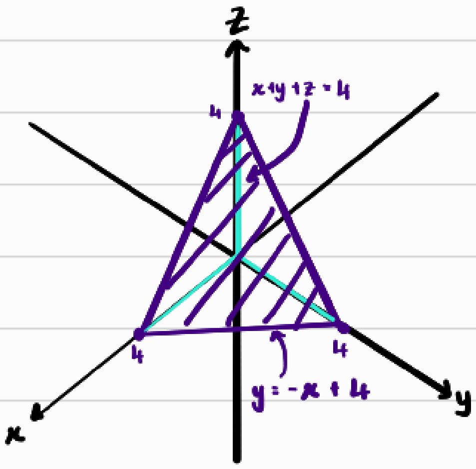
Application : Volume , mass & centre of mass

Example 1 :

Properties of tetrahedron , $T = \{(x,y,z) \in \mathbb{R}^3 \mid x \geq 0, y \geq 0, z \geq 0, x+y+z \leq 4\}$

with density $\rho(x,y,z) = 3$

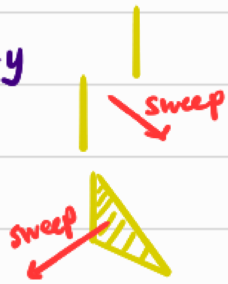




$0 \leq z \leq 4-x-y$

$0 \leq y \leq 4-x$

$0 \leq x \leq 4$



V $= \iiint_T 1 \, dV$

$= \int_0^4 \int_0^{4-x} \left[\int_0^{4-x-y} 1 \, dz \right] dy \, dx$

$= \int_0^4 \int_0^{4-x} (4-x-y) \, dy \, dx$

$= \int_0^4 \frac{1}{2}(16 - 8x + x^2) \, dx$

$= 32/3$ #

m $= \iiint_T \rho \, dV$

$= \int_0^4 \int_0^{4-x} \left[\int_0^{4-x-y} 3 \, dz \right] dy \, dx$

$= 3 \left[\int_0^4 \int_0^{4-x} \int_0^{4-x-y} dz \, dy \, dx \right.$

$= 3 \, (32/3)$

$= 32$ #

Centre of mass

$\bar{x} = \frac{1}{m} \iiint_T x\rho \, dV$

$= \frac{1}{32} \int_0^4 \int_0^{4-x} \int_0^{4-x-y} 3x \, dz \, dy \, dx$

$\vdots$

$= 1$ #

$\bar{y} = \frac{1}{m} \iiint_T x\rho \, dV$

$= \frac{1}{32} \int_0^4 \int_0^{4-x} \int_0^{4-x-y} 3y \, dz \, dy \, dx$

$\vdots$

$= 1$ #

$\bar{z} = \frac{1}{m} \iiint_T x\rho \, dV$

$= \frac{1}{32} \int_0^4 \int_0^{4-x} \int_0^{4-x-y} 3z \, dz \, dy \, dx$

$\vdots$

$= 1$ #

Change of coordinates

Same concept as double integrals [compensating for the stretch in new coordinate system], but for an extra dimension. Since there are 3 variables, the Jacobian matrix becomes 3 × 3.

$$J(u,v,w) = \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{bmatrix}$$

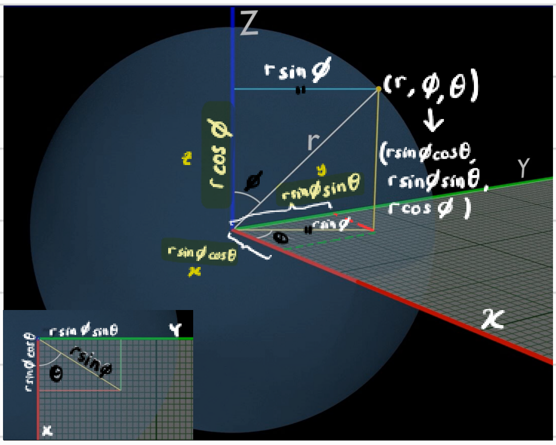
Take Note

The mathematical convention for spherical is (r, θ, ϕ)

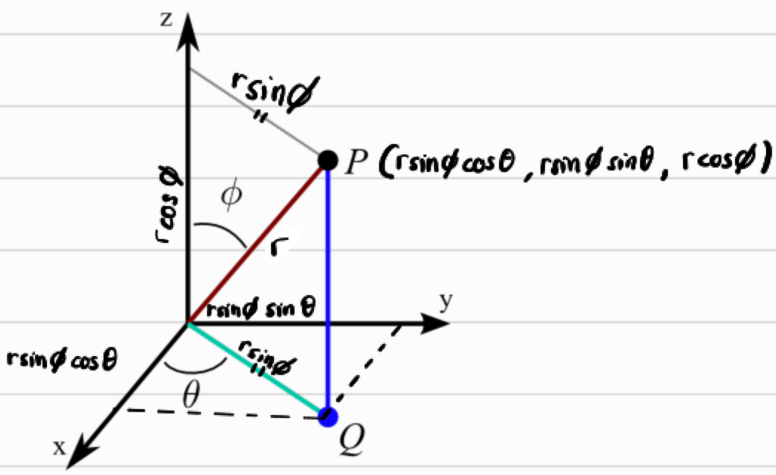
θ being angle around z-axis

ϕ being angle b/w z-axis

Spherical coordinates [3D polar coordinate]



https://youtu.be/FDyenWWlPdU?si=Obx5DCRA8eHxUKHK



$\therefore \quad x = r\sin\phi\cos\theta$

$\quad\quad y = r\sin\phi\sin\theta$

$\quad\quad z = r\cos\phi$

θ : Angle w/ the x-axis

ϕ : Angle w/ the z-axis

Take note:

$\phi = 0$: +ve z-axis

$\phi = \pi/2$: $z = 0$ plane

$\phi = \pi$: -ve z-axis

$r = \sqrt{x^2 + y^2 + z^2}$

$\phi = \cos^{-1}\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)$

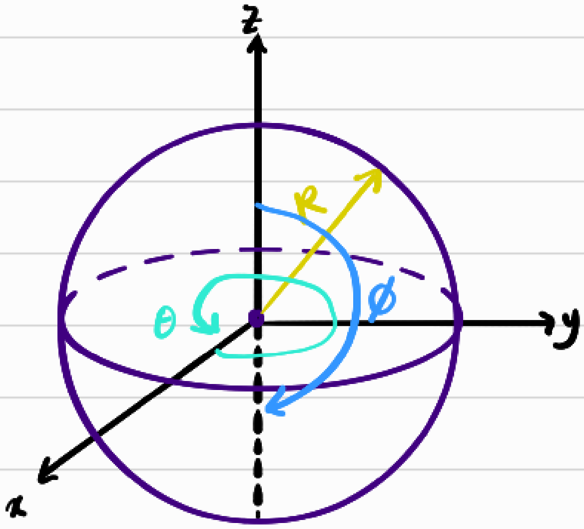
$\theta = \tan^{-1}\left(\frac{y}{x}\right)$

Jacobian is given by,

$$J(r,\phi,\theta) = \det \begin{bmatrix} \cos\theta\sin\phi & -r\sin\theta\sin\phi & r\cos\theta\cos\phi \\ \sin\theta\sin\phi & r\cos\theta\sin\phi & r\sin\theta\cos\phi \\ \cos\phi & 0 & -r\sin\phi \end{bmatrix} = r^2\sin\phi$$

Example 1

Find the Volume of a ball of radius R units.



ball B

Polar	Cartesian
$0 \le r \le R$	$-\sqrt{R^2-x^2-y^2} \le z \le \sqrt{R^2-x^2-y^2}$
$0 \le \theta \le 2\pi$	$-\sqrt{R^2-x^2} \le y \le \sqrt{R^2-x^2}$
$0 \le \phi \le \pi$	$-R \le x \le R$

Horrible to integrate

$$V = \iiint_B 1 \, dx\, dy\, dz$$

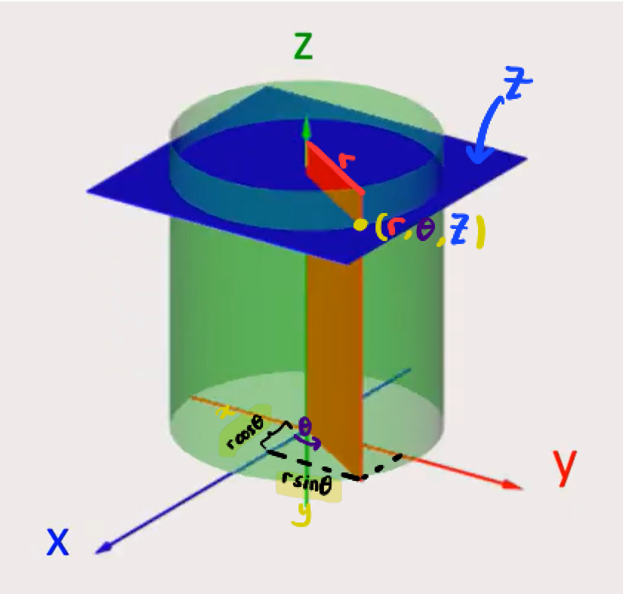
$$= \iiint_{B'} 1 \cdot \underbrace{|J(r,\theta,\phi)|}_{r^2 \sin\phi} \, dr\, d\theta\, d\phi$$

$$= \int_0^{\pi} \int_0^{2\pi} \int_0^{R} r^2 \sin\phi \, dr\, d\theta\, d\phi$$

$$\vdots$$

$$= \frac{4}{3}\pi R^3 \quad \#$$

Cyllindrical Coordinate



https://youtu.be/EthQB3325GM?si=gz-anVteOgaU6LcY

$x = r\cos\theta$	$r = \sqrt{x^2+y^2}$
$y = r\sin\theta$	$\theta = \tan^{-1}(y/x)$
$z = Z$	$Z = z$

The Jacobian is given by

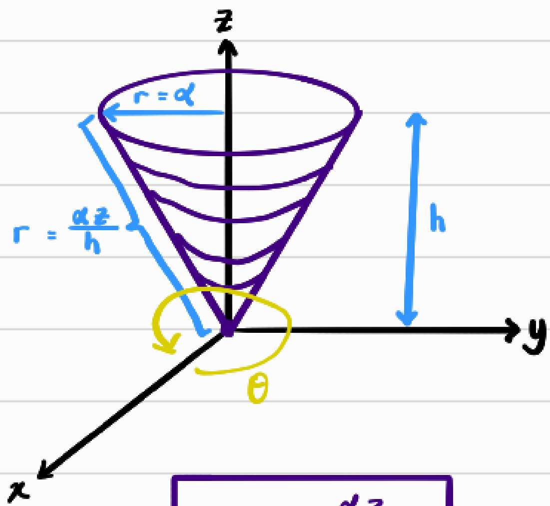
$$\det\begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \cancel{\frac{\partial x}{\partial Z}}\,0 \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \cancel{\frac{\partial y}{\partial Z}}\,0 \\ \cancel{\frac{\partial z}{\partial r}}\,0 & \cancel{\frac{\partial z}{\partial \theta}}\,0 & \cancel{\frac{\partial z}{\partial Z}}\,1 \end{bmatrix}$$

x not dependent on Z
y not dependent on Z
z not dependent on r
z not dependent on θ

$$J(r,\theta,Z) = \det\begin{bmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = r(\cos^2\theta + \sin^2\theta) = r \ \#$$

Example 1

Find the Volume of a cone of radius α units and height h units



$$\boxed{\begin{aligned} 0 &\le r \le \frac{\alpha z}{h} \\ 0 &\le \theta \le 2\pi \\ 0 &\le z \le h \end{aligned}}$$

Jacobian

$$V = \int_0^h \int_0^{2\pi} \int_0^{\alpha z/h} 1 \; r \; dr\, d\theta\, dZ$$

$$\vdots$$

$$= \frac{\pi\alpha^3 h}{3} \ \#$$